Invoice pipeline

First day in Moty's cost department. Nobody kept the invoices.

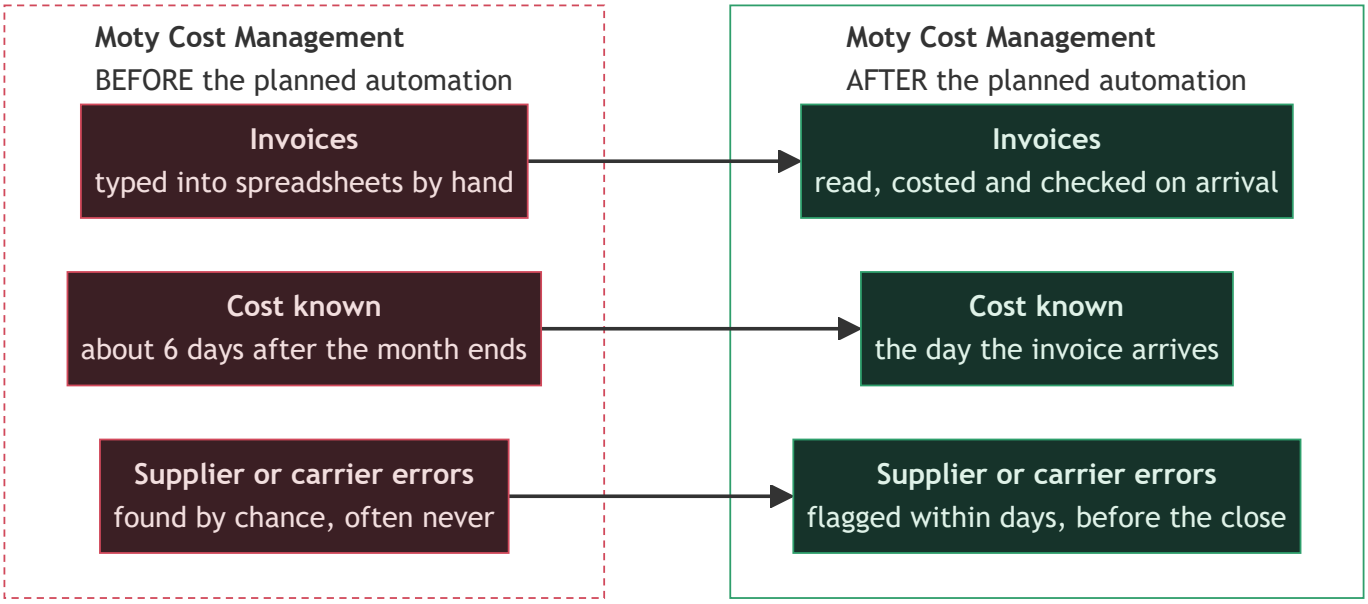
Moty imports cosmetics packaging from Asia: bottles, caps, pumps, labels. Its plant is in Louveira, a real city in São Paulo state; Moty itself is invented. Since 1 January, factories, shipping lines, forwarders, a customs broker, a terminal and a truck company have billed it, *and none of it was filed*.

You work in **Production and Logistics Cost Management**. Your job is to know **what every part cost to land at the plant in Louveira**, by carrier, by operation and by product, *and to answer management the day they ask*.

The department does not pay invoices and does not choose what is bought. **It only works out what things cost.**

So the first thing you do is **e-mail every sender: send us every document issued to Moty since 1 January.** They reply, *each in its own way*, a few PDFs per mail.

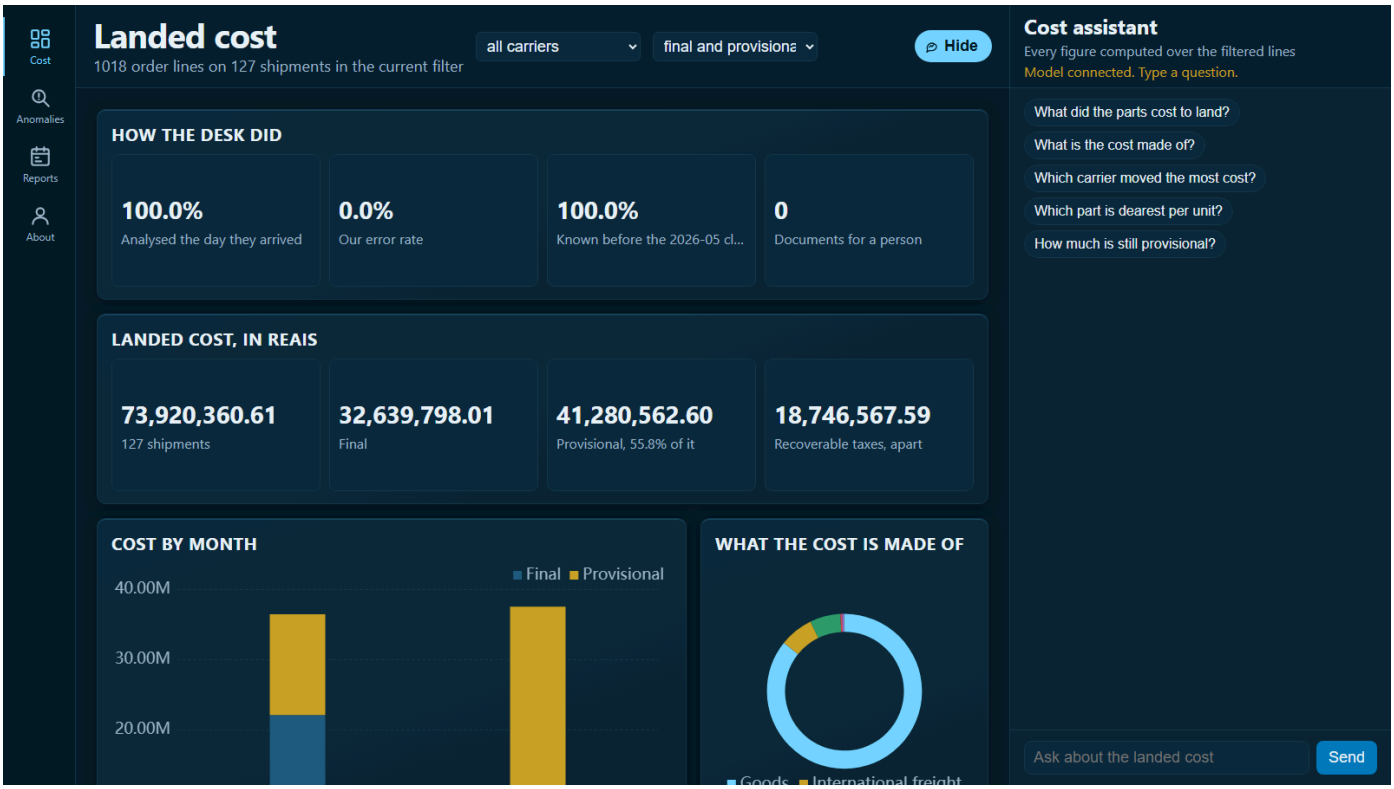
The scenario, before and after:



Why it matters:

- Closing the books takes a typical company **about 6 days** a month [1].
- **3 to 6 of every 100** freight invoices have an **error** [2].

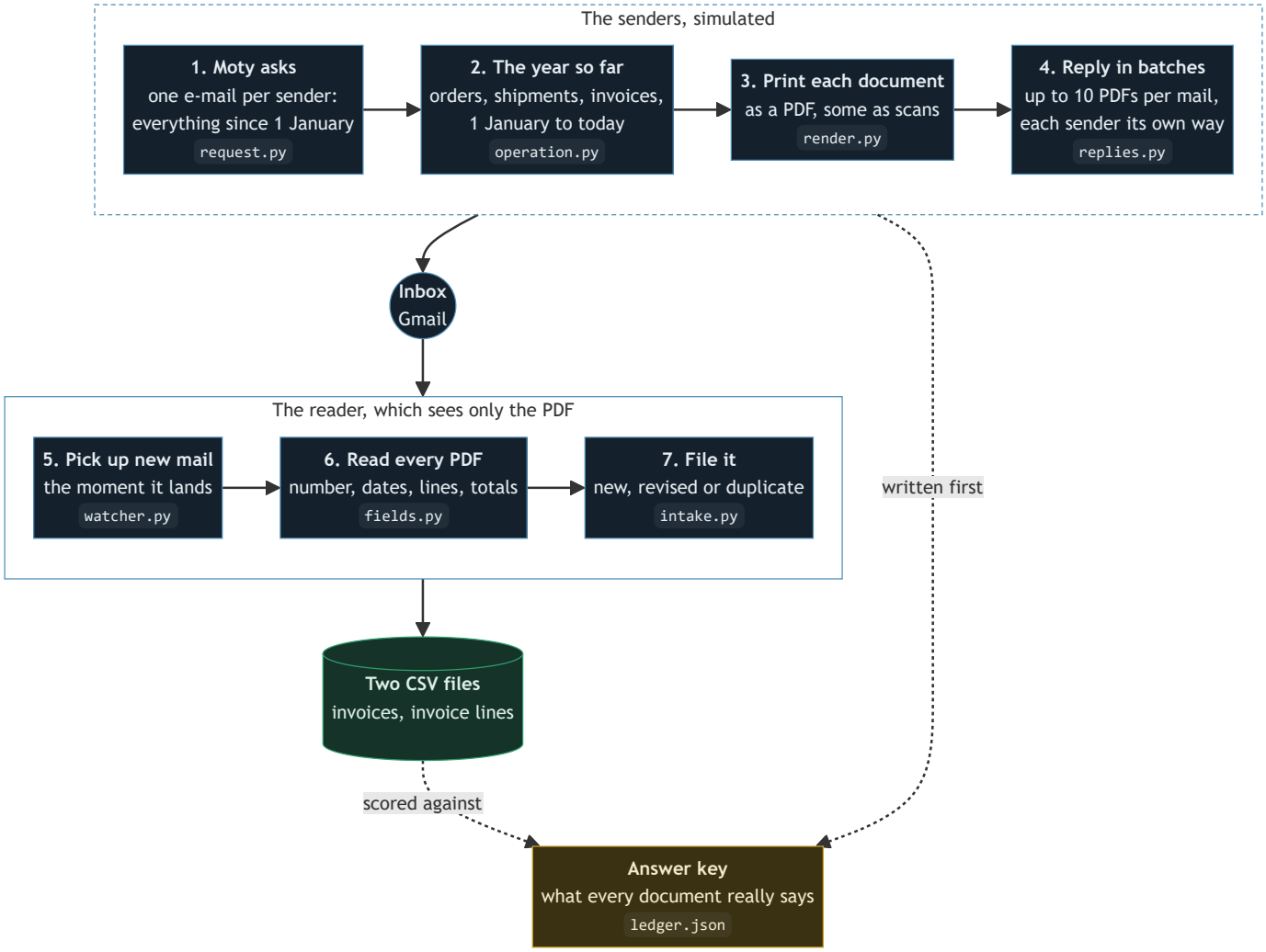
The numbers end up in the landed cost dashboard, *where management asks in plain words and every figure is computed, never guessed:*



Its code: the dashboard repository

The shape of it

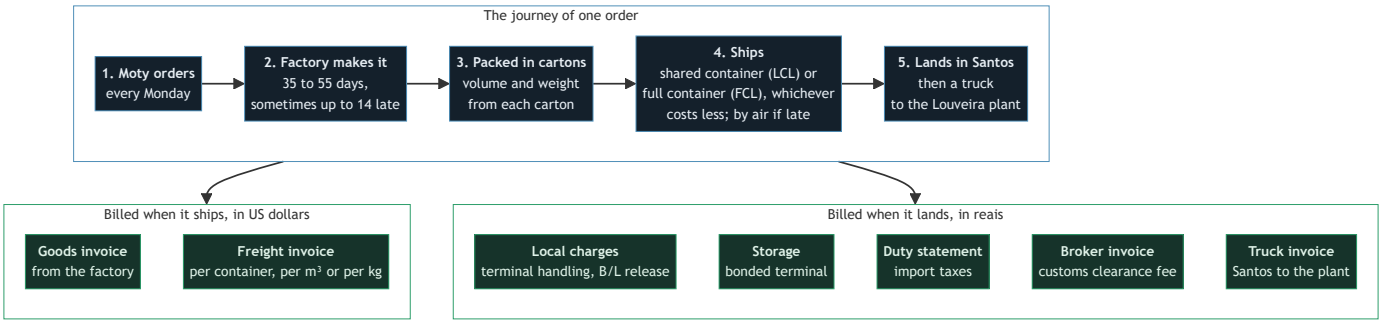
From the request to the cost report. *The answer key is written first and kept away from the reader.*



Nothing is dated after today. Each later run brings only what was issued since the last one.

Where an invoice comes from

Every document comes from an event in the business: an order, a shipment, an arrival.



- **A small order ships shared (LCL)**, charged per m^3 or per tonne, whichever is greater.
- **A big one fills its own container (FCL)**, one price per box, however full.
- *Each shipment compares the two, all charges included, and takes the cheaper*, as a forwarder quotes both. With 2026 rates the switch falls near **46 to 60 m^3**, so Moty ships mostly LCL.

- A late order flies 55% of the time if it is small enough, charged per kilo or by volume, whichever is greater.

What the reader is up against

- **A backlog, not a drip.** Months of documents arrive at once, in replies of up to 10 PDFs, each sender batching, grouping and wording them its own way.
- **Thirteen issuers, each printing its own way.** The same day arrives as `2026-04-11`, `04/11/2026` or `11/04/2026`; amounts as `1,234.56` or `1.234,56`.
- **Two senders scan.** The Indian label printer sends most of its invoices as cheap black-and-white scans, half of them askew; the broker scans a signed copy now and then. They are read with OCR. If the lines do not add up to the total, *a person gets the row*; the reader never guesses.
- **Resends, revisions and credit notes**, as in any real inbox.
- **Ten kinds of error, planted on purpose** for a model to catch later. Each has an *honest twin* that looks the same on the page.

And once read, **every part is costed in reais**: freight, local charges, storage, import duty, broker and truck, each split the way it is charged; the recoverable taxes kept apart.

How well it reads

The reader is scored against the answer key on every run. The latest figures, by scan quality: the reference.

Set up

You need Python 3.10 or newer.

```
git clone https://github.com/RafaelTavares98/invoice-pipeline
cd invoice-pipeline
python -m venv .venv
.venv/Scripts/activate        # .venv/bin/activate on Linux and macOS
pip install -r requirements.txt
```

To read scanned invoices, also install Tesseract (`winget install UB-Mannheim.TesseractOCR` on Windows, `sudo apt-get install tesseract-ocr` on Ubuntu). Without it, scans are filed for a person.

Or with Docker, no Python needed:

```
docker build -t invoice-pipeline .
docker run --rm invoice-pipeline
```

Check that it works

```
python run.py smoke
```

You should see:

```
smoke: 1 of 1 read correctly; scans 0 of 0; in ...\Invoices\samples\single
```

One reply came back with one invoice, and the reader filed it and checked it against its answer, in about a second. Everything it wrote is in `Invoices/samples/single/`.

Go further

- **Tutorial**: send the requests, receive the backlog, read it, score the reader and cost every part, step by step.
- **Reference**: every command, file, column, charge and rate.

Sources

1. APQC, "Metric of the Month: Cycle Time for Monthly Close", CFO.com. Median 6.4 calendar days; top quarter 4.8. https://www.cfo.com/news/metric-of-the-month-cycle-time-for-monthly-close/659297/
2. Transportation Insight, "Freight Audit and Payment: Protecting Profit in Soft Rates". A typical audit finds errors on 3 to 6% of invoices. https://transportationinsight.com/resources/freight-audit-payment-protect-profit-soft-rates/

The freight rates and charges behind the model are sourced in the reference.